AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 19 · STUDENT LESSON

Similarity

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

State the features. Then compare the scores.

Score Basic1, inspect tied Craft2 results, add a candidate and compare a new book task.

Keep one habit: Keep records fixed → name the weights → return the stated ties and limit suitability claims.

MAIN LESSON · SECTION 1

Similar in which selected ways?

> Calculate the declared score, handle ties explicitly and keep suitability requirements separate.

Start with a changed ranking

An item can win under one similarity rule and lose under another without changing at all. State the features, weights and candidate set before interpreting its score.

Retrieve the score-meaning question

A system gives an item two matching-feature points. Does that mean a two-thirds chance that the item is suitable? What must you read first?

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

By the end, you can...

- I can calculate **similarity** using explicitly selected features and a stated scoring rule.
- I can explain how a change in feature **weight**s can change the highest scoring item.
- I can handle ties and missing features without claiming that a **similarity** score proves suitability.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

Without opening the old lesson, explain one key idea from Confidence Is Not Correctness. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Boundaries and 'Not Sure'. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Learning from Examples. Give an example and a mistake that the idea helps you avoid.

Recommendations often begin by comparing selected traits, not whole objects.



LEARNING TARGET A reader wants mystery and short books. A has both traits, B only mystery, C only short. Score one point per matching trait. The reader now says short is essential. Explain how the decision should change.

Learner name:

Readiness for the connected explanation

This week’s end goal

By the end, I can calculate **similarity** under a declared feature rule and explain a changed ranking.

MAIN LESSON · SECTION 2

Define Basic1 before scoring the cards

> A similarity score belongs to its chosen features and comparison rule.

State the purpose and selected features

Two objects can be alike in colour and different in shape. A computer comparison needs a precise answer to similar in what way. Select features, state how their values are compared, and define what a larger score means.

The result belongs to that rule; it is not a universal judgment about the objects.

Read Basic1's full target and candidate records

Our target card is red, round and smooth. Three **candidate** cards are A: red, round, rough; B: blue, square, smooth; C: red, square, rough.

First give one point for each feature that exactly matches the target and zero for each known mismatch. Add the points. Colour, shape and texture are the only selected features.

Keep unknown separate from mismatch

This paper exercise uses complete category values with agreed spelling. Real records can be missing or uncertain. Do not silently treat an unknown texture as a known mismatch or copy a value from the target.

First identify whether the information supports the comparison.

Name the comparison version

- Call this first comparison Basic1. Name its target, the three feature fields and the one-point rule. Which missing field would stop a complete comparison under our classroom policy?

▼ Earlier wording, alternatives and picture transcripts

PICTURE STORY

Look Closely Before You Explain

Pictures give clues; **evidence** tells us what the clues mean.



MYSTERY A book can be nearest to another by reading level, topic, length, or mood; the chosen features and scales determine the neighbor.

What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first **prediction** and state the clue that supports it.

My **observation**s, **prediction**, and **evidence**:

GOOD THINKING A **prediction** can change when stronger **evidence** appears.

A closer explanation

Show each match contribution before adding

> For Basic1, larger exact-match totals mean more matches under the declared rule.

Read the three candidate contributions

Three features contribute to a score

CANDIDATE	MATCHES WITH TARGET	ONE POINT PER MATCH
A: red round rough	Colour and shape	2 points
B: blue square smooth	Texture only	1 point
C: red square rough	Colour only	1 point

The target is red, round and smooth. A has the largest score under equal weights. These points count selected feature matches; they are not probabilities or evidence that A suits every purpose.

Check all feature-level points

A receives 1 for colour, 1 for shape and 0 for texture: total 2. B receives 0, 0 and 1: total 1. C receives 1, 0 and 0: total 1. Record the contributions, not just the totals, so another learner can check the computation.

Limit the original winning claim

A is most similar among these three **candidate**s under this rule. That sentence names the **candidate** set and the rule.

It does not say A is identical to the target, has the right texture or will be the best choice for a task that needs smooth material.

Explain one score

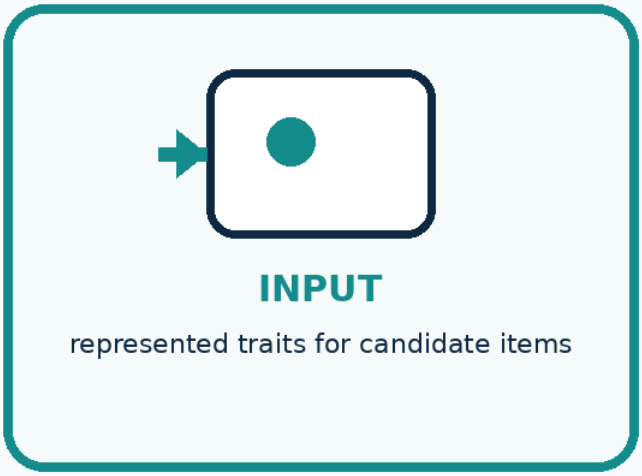
- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

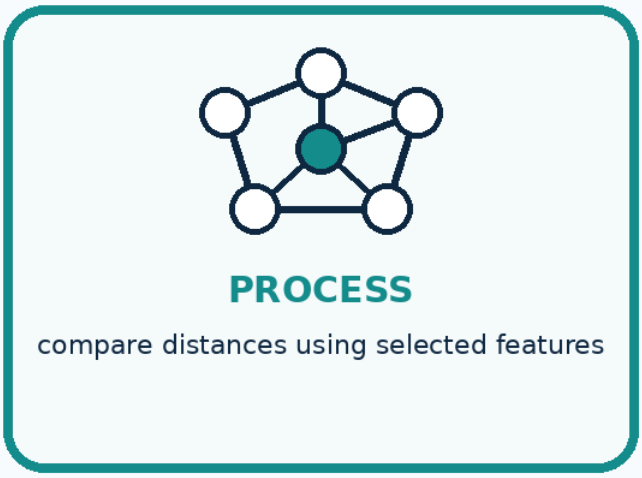
▼ Earlier wording, alternatives and picture transcripts

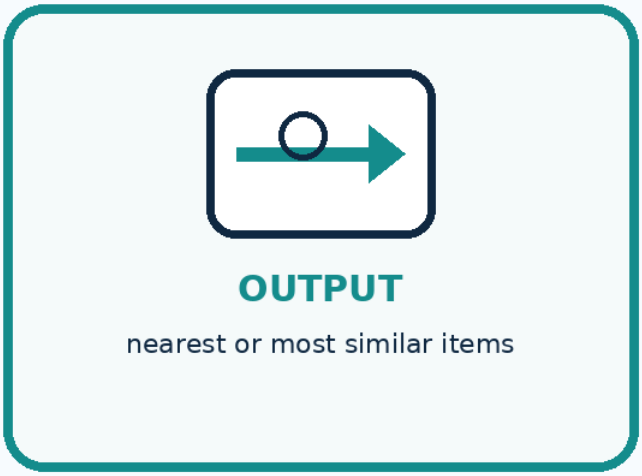
VISUAL **MECHANISM**

Input → Process → Output

Follow the information instead of imagining magic inside the machine.







INPUT represented traits for **candidate** items

PROCESS compare **distance**s using selected features

OUTPUT nearest or most similar items

CORE IDEA Recommendations often begin by comparing selected traits, not whole objects.

UNDER THE HOOD **Similarity** is a designed measure over represented features. **Distance** changes with feature selection, scaling, **weight**ing, and missing values. Nearest does not mean identical, best, or fair.

Draw a fresh example with all three stages:

A closer explanation

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Compare selected traits and recommendation clues

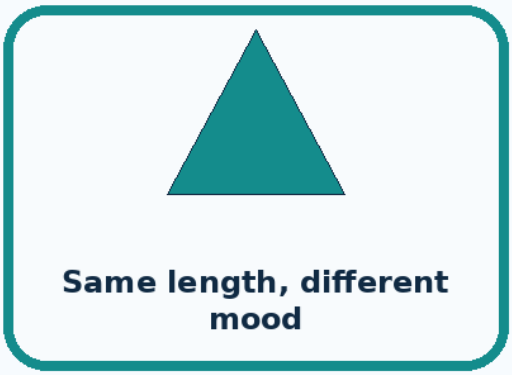
> Ask what the features represent and which task requirements a score might leave out.

Pictures paired with their evidence



The case to investigate

CASE 1 — Same topic, different level Clue: close on topic. Meaning: far on reading level.



The case to investigate

CASE 2 — Same length, different mood Clue: close on length. Meaning: far on mood.



The case to investigate

CASE 3 — **Weight**ed topic Clue: topic counts more. Meaning: neighbor changes.



The case to investigate

CASE 4 — Missing feature Clue: **distance** incomplete. Meaning: need policy.

Discuss the confusing pair

- Which two pictures are easiest to confuse? What **evidence** separates them?

Keep the limit in view

BOUNDARY **Similarity** is a tool designed for a purpose, not a universal property.

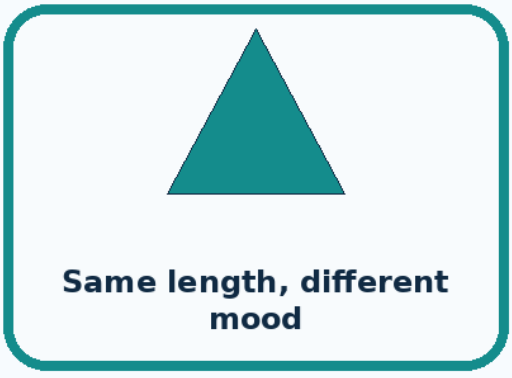
▼ Earlier wording, alternatives and picture transcripts

PICTURE GALLERY

Four Cases, Four Clues

Use the visual cue, then read the **evidence** carefully.









CASE 1 — Same topic, different level Clue: close on topic. Meaning: far on reading level.

CASE 2 — Same length, different mood Clue: close on length. Meaning: far on mood.

CASE 3 — **Weight**ed topic Clue: topic counts more. Meaning: neighbor changes.

CASE 4 — Missing feature Clue: **distance** incomplete. Meaning: need policy.

Change one thing in the pictured case

Change: Keep original **weight**s but change the target texture to rough.

Prediction: C scores 3; A and B each score 1.

Reason: C now matches all three target features; the others lose their smooth-texture match.

A declared comparison can have more than one winner

> Keep the Craft2 dataset and each weight version explicit.

Craft2’s target, records and two declared rules

This separate worked dataset is Craft2. Its A, B and C records are printed below; use these records for this case. Return every highest scorer under each stated comparison.

A fictional craft-card matcher compares each **candidate** with a target.

Target: red, round, smooth. A: red,square,smooth. B: blue,round,smooth. C: red,round,rough.

Initially each exact **feature match** scores 1; return all highest scorers.

For the second comparison only, colour and shape matches score 1 each and a texture match scores 3.

Compare the tied score patterns

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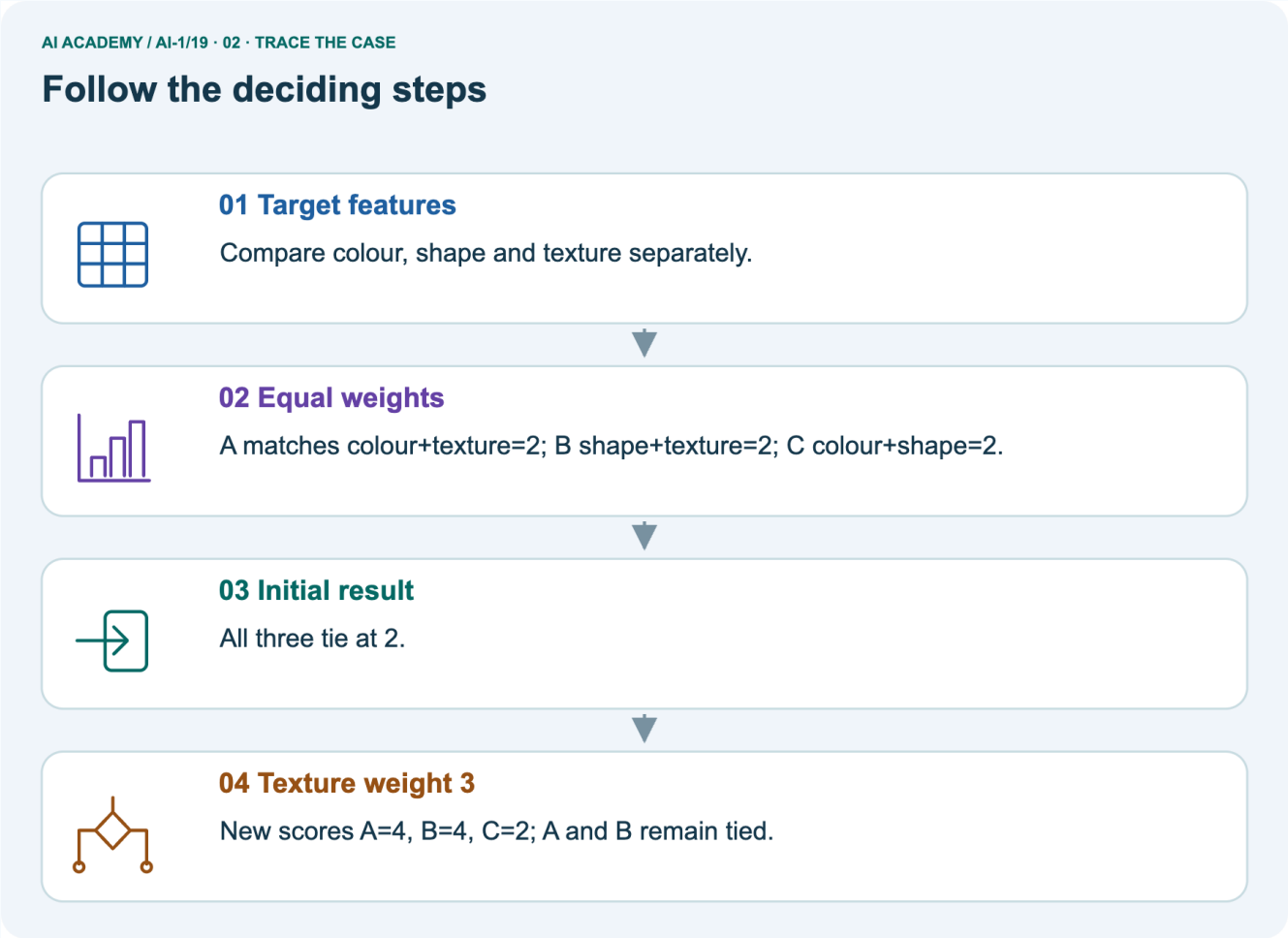
Declared weights change similarity

	All weights 1	Texture weight 3
A	2	4
B	2	4
C	2	2

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

Trace both weighted results



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain the tie policy and changed priorities

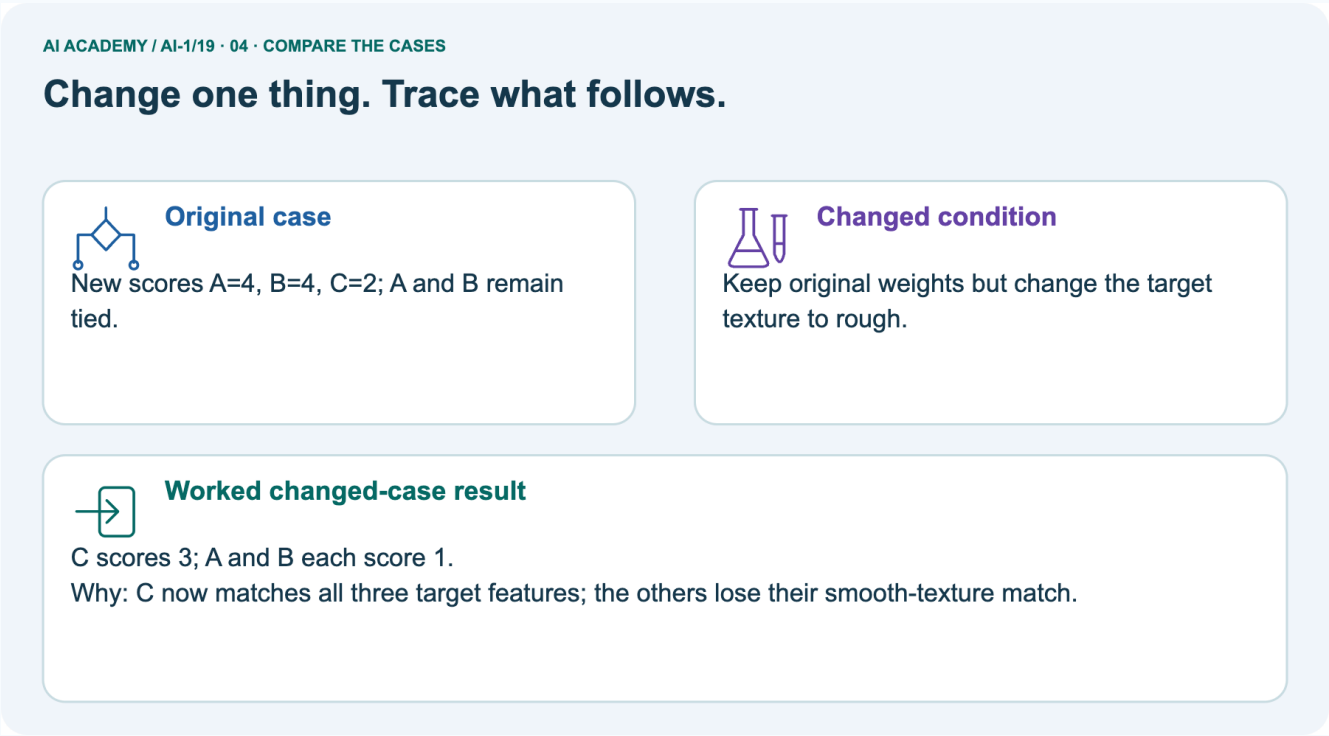
Similarity is defined by the chosen representation and **weight**s. The cards did not change when texture gained **weight**; the comparison rule did. Two high scores need not mean two identical cards.

A complete rule also says what to do with tied scores, rather than choosing whichever appears first by accident.

Predict after changing the target texture

- Before checking, use Craft2′s original one-point weights but change the target texture from smooth to rough. Leave its colour and shape unchanged. Recalculate every contribution and return all highest scorers.

▼ Check the changed-target scores



Follow the labelled picture with the matching explanation. Open the vector diagram

Limit similarity to represented properties

A high feature-match score does not establish usefulness or suitability beyond the features represented.

Explain C’s changed score

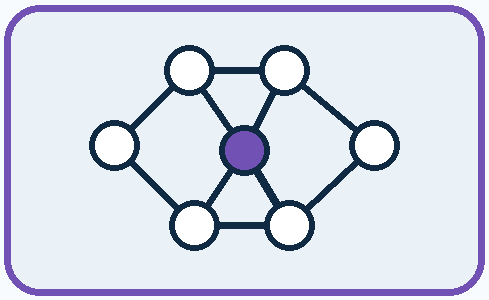
- Read the labelled picture: Point to the matches supporting C’s new score and explain why there is no tie.

▼ Earlier wording, alternatives and picture transcripts

WORKED EXAMPLE

Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable **evidence**.



CASE A book can be nearest to another by reading level, topic, length, or mood; the chosen features and scales determine the neighbor.

STEP 1 — OBSERVE represented traits for **candidate** items

STEP 2 — TRACE compare **distance**s using selected features

STEP 3 — CONCLUDE nearest or most similar items

MODEL REASONING Plot items consistently, define how each axis is measured, and compare **distance**s. A different axis or **weight** can change the nearest item without any item moving in the real world.

DO NOT OVERCLAIM **Similarity** is a tool designed for a purpose, not a universal property.

What new **evidence** could change this answer?

A closer explanation

Candidate	Colour and shape points	Texture worth 3 points	New total
A	1 + 1	0	2
B	0 + 0	3	3
C	1 + 0	0	1

Suppose texture matters more for the stated comparison. Keep colour and shape worth one point each, but give three points for a texture match. B now scores 3 and becomes the unique highest scoring **candidate**.

No card changed; the rule changed. Record both versions rather than overwriting the first result.

Weights express priorities in this designed rule. They need a reason connected to the task and should be set before inspecting the final evaluation results. A convenient winner does not justify a **weight**.

If smooth texture is required without exception, use a requirement check rather than hoping a **weight**ed score will always enforce it.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Similarity

The facts we will use

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Declared weights change similarity

	All weights 1	Texture weight 3
A	2	4
B	2	4
C	2	2

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/19

Similarity

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Target features

Compare colour, shape and texture separately.



2. Equal weights

A matches colour+texture=2; B shape+texture=2; C colour+shape=2.



3. Initial result

All three tie at 2.



4. Texture weight 3

New scores A=4, B=4, C=2; A and B remain tied.

CHANGE ONE THING

Keep original weights but change the target texture to rough.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Target features	Compare colour, shape and texture separately.
2. Equal weights	A matches colour+texture=2; B shape+texture=2; C colour+shape=2.
3. Initial result	All three tie at 2.
4. Texture weight 3	New scores A=4, B=4, C=2; A and B remain tied.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

What this example does not establish

Change one thing: Keep original **weight**s but change the target texture to rough. Predict the result before checking the explanation. What remains the same?

Practise: add a candidate and return every tied winner

> First use Basic1's one-point rule, then make texture worth 3 as a separately named comparison.

Basic1 and the new D record

Return to Basic1 for this practice: target red, round, smooth; A red, round, rough; B blue, square, smooth; C red, square, rough. The additional D is blue, round, smooth. Keep these records separate from Craft2's records.

Add **candidate** D: blue, round, smooth. Under the original one-point rule, D scores 2, tying A. Our policy is to return every **candidate** with the largest score, so the result is A and D.

Returning only the first row would silently introduce a different policy.

> **Using A, B, C and D, calculate the four totals under the original rule and then under texture worth 3. Return every highest scoring candidate each time. Explain which matches caused the change and why neither result establishes a probability of suitability**

Pause for a missing comparison value

If a required comparison value is missing, our classroom policy pauses the final ranking and requests the missing value. Retain the known contributions and the reason for the pause.

A different policy could be designed, but it would need to be explicit and checked for its effects. Unknown and different are separate states.

Use W2 to explain the changed winner

Use workbook W2 for the connected example "**Similarity** and selected features". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Challenge a universal meaning of similar

- Use the worked case for **Similarity** and selected features. Proposed explanation: "Similar has one objective meaning for every task". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Return to Craft2's changed-target explanation

Point to the matches supporting C's new score and explain why there is no tie.

> **My explanation using a pictured step and the given facts**

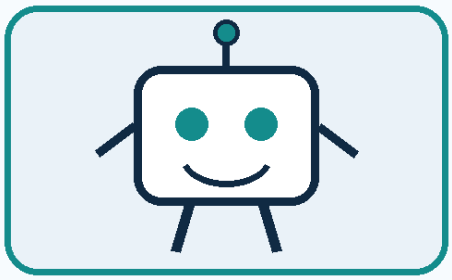
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Earlier wording, alternatives and picture transcripts

GUIDED PRACTICE

Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: Recommendations often begin by comparing selected traits, not whole objects.

Use a fresh example to explain **similarity**.

Trace a new **input** → **process** → **output** example.

Find and repair the error in: Two items have one permanent **similarity** score.

Invent one edge case and name the safe response.

PARTNER CHECK Point to the **evidence**. Do not accept an answer that only repeats the vocabulary word.

A closer explanation

Practise with less help: Explain a changed winner

Check your reasoning

Explain the picture

Qualify a claim about the “closest” item

> Use the rule, candidate set and tie policy when explaining a winner.

Ask which features were left out

Feature selection leaves information out. A card comparison ignores size and material unless those are included. A book comparison using topic and format may ignore language or reading difficulty.

More features are not automatically better: irrelevant or repeatedly copied features can change a score without improving the decision.

State the full scoring and missing-value policy

A useful report states the purpose, **candidate**s, selected features, **weight**s, missing-value policy and tie policy. Check the proposed method on relevant cases that include ties and imperfect records.

If it supports a recommendation, keep the explanation limited to what was actually compared.

Repair the objective-winner claim after adding D

- CLAIM A The nearest item is automatically the best recommendation.
- CLAIM B Two items have one permanent **similarity** score.

STRONG ANSWER FRAME I claim _____ because the **evidence** shows _____. The **process** is _____. One limit is _____.

> **A learner says A is objectively the closest card because it won the original comparison. Use the changed texture weight and the added candidate D to explain why the claim needs qualification**

▼ Earlier wording, alternatives and picture transcripts

COMPARE

Same Result, Different Reason

A comparison helps reveal the true concept boundary.

Case	Evidence	What it shows
Same topic, different level	close on topic	far on reading level
Same length, different mood	close on length	far on mood
Weighted topic	topic counts more	neighbor changes
Missing feature	distance incomplete	need policy

Repair two tempting claims

Repair:

Repair:

A closer explanation

MAIN LESSON · SECTION 9

Investigate two different trait-map neighbours

> Use a fully specified map, two declared priorities and an explicit tie and missing-value policy.

Your map and comparison rules

Draw a trait map with rows red/blue and columns round/square. The target is red,round. Put A=red,square; B=blue,round; C=blue,square in their cells. Start with one point per exact feature match, returning all highest totals. Then compare two policies declared before

scoring: colour-priority gives colour 2 and shape 1; shape-priority gives colour 1 and shape 2.

Predict, score and compare priorities

1. Predict all totals and winners under the three stated policies before checking with a partner. Explain how A and B become different highest scorers under the two priorities. For an added red card whose shape is unreadable, retain the known colour contribution and request its shape before final ranking. These points are feature-match scores, not chances of suitability.

2. Keep a normal case and an edge case clearly separated.

3. Record what happened—even when it does not match your **prediction**.

4. Explain the result, one limitation, and the next useful test.

Keep rule versions and limits

Case	Prediction	Observed result	What it means
Normal			
Edge			

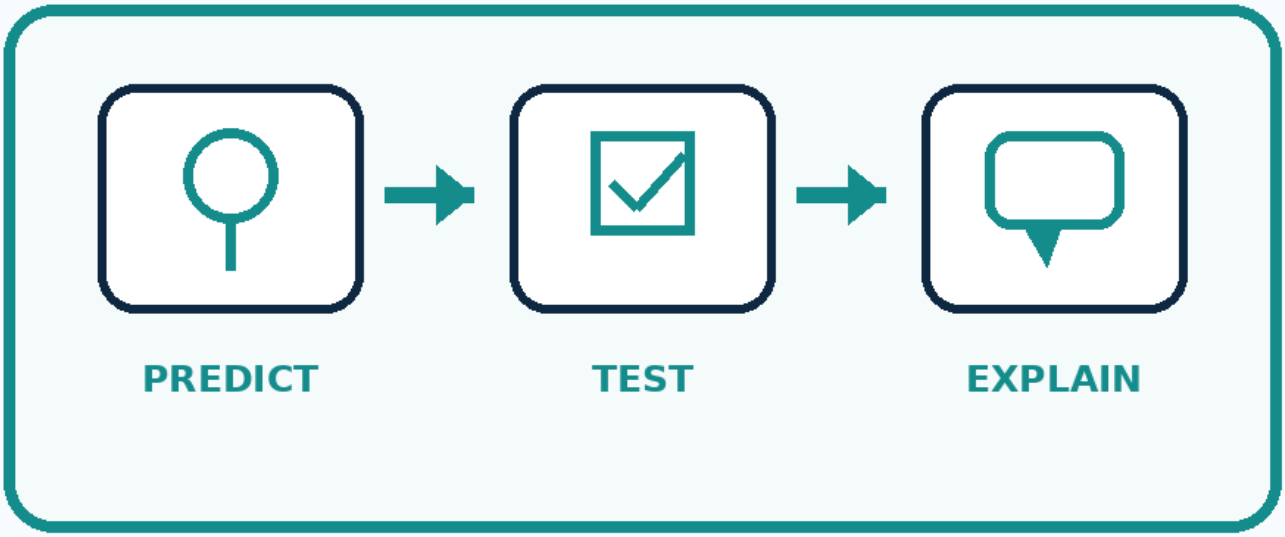
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

EXPLORER LAB

Trait Map Neighbors

Predict first, test honestly, and explain what the **evidence** means.



State your **prediction** before the result is known.

Complete the task: Create a trait map and find two different 'nearest' items.

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: compare the new book records

> Use the topic, format and language fields and return all tied highest scorers before considering a hard requirement.

The new target, records and weight change

A target book has topic space, format comic and language English. **Candidate**s are J: space, prose, English; K: animals, comic, English; L: space, comic, Spanish. Give 1 point per exact match and return all tied highest scorers.

Then make topic worth 3 points while keeping the other **weight**s at 1. Report both sets of totals and winners. Explain why a winner may still fail a requirement to read English.

> **My independent explanation, trace and limits**

Keep one first attempt

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback afterwards

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Transfer the comparison to books

Week 19 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Compare books using topic only. Book A is about space with a red cover; B is about space with a blue cover; C is about dogs with a red cover. Which matches A best?

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Choose two useful item traits for **similarity**-based sorting and explain why each deserves its **weight**.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Transfer practice from the original programme

A reader wants mystery and short books. A has both traits, B only mystery, C only short. Score one point per matching trait. The reader now says short is essential. Explain how the decision should change.

Explain the mechanism, evidence and one limit.

Use feedback to improve

MAIN LESSON · SECTION 11

Declare your sorter’s similarity rule before using it

> Give each feature and weight a task-related reason and state tie behaviour.

This week’s project decision

THIS WEEK'S CHECKPOINT Choose two useful item traits for **similarity**-based sorting and explain why each deserves its **weight**.

Declare sorter **similarity** features, **weight**s and tie behavior before using scores to explain recommendations.

Plan traits, scoring checks and review

- What real classroom need does this serve?
- What information is necessary—and what should stay private?
- What happens inside the **process**?
- What normal and edge cases will you test?

- When should it say not sure or ask a person?

Keep task-specific similarity limited

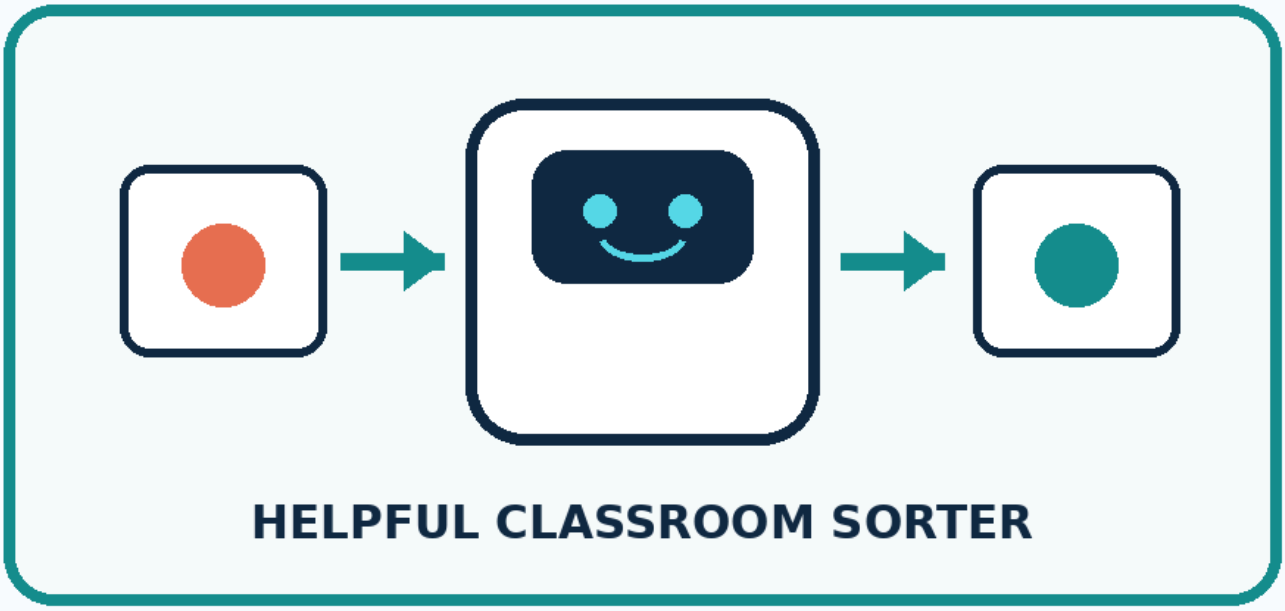
RESPONSIBLE DESIGN **Similarity** is a tool designed for a purpose, not a universal property.

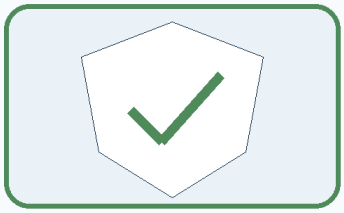
▼ **Earlier wording, alternatives and picture transcripts**

PROJECT

Helpful Classroom Sorter

Build one coherent system across the year, one **evidence**-based decision at a time.





My project decision:

Evidence for your project

MAIN LESSON · SECTION 16

Finish: explain the score and its limits

> Close your notes and state the feature contributions, rule version, tie result and missing requirement.

Your exit explanation

- Close your notes. Explain this goal in your own words: Calculate **similarity** using explicitly selected features and a stated scoring rule

> **My deciding step, evidence and one remaining question**

Choose support and revisit later

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	Meaning	My example
similarity	degree of likeness under chosen traits	
distance	numeric or ordered difference between representations	
weight	importance assigned to a feature	
nearest neighbor	item with smallest chosen distance	

Terms for the connected explanation

Word	Meaning
Feature match	Agreement between two recorded values for a selected feature
Tie	Two or more candidates sharing the same score
Candidate	An item being considered in a comparison

Repair a claim using a word

- CLAIM TO REPAIR The nearest item is automatically the best recommendation.
- Use one vocabulary word and one piece of **evidence** to repair it:

▼ **Earlier wording, alternatives and picture transcripts**

VOCABULARY

Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.









Misconception detective

WORD POWER A strong definition helps you decide something in a new case.

Words used in the closer explanation

▼ **Choose a stronger thinking challenge**

REFERENCE · SECTION 10

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

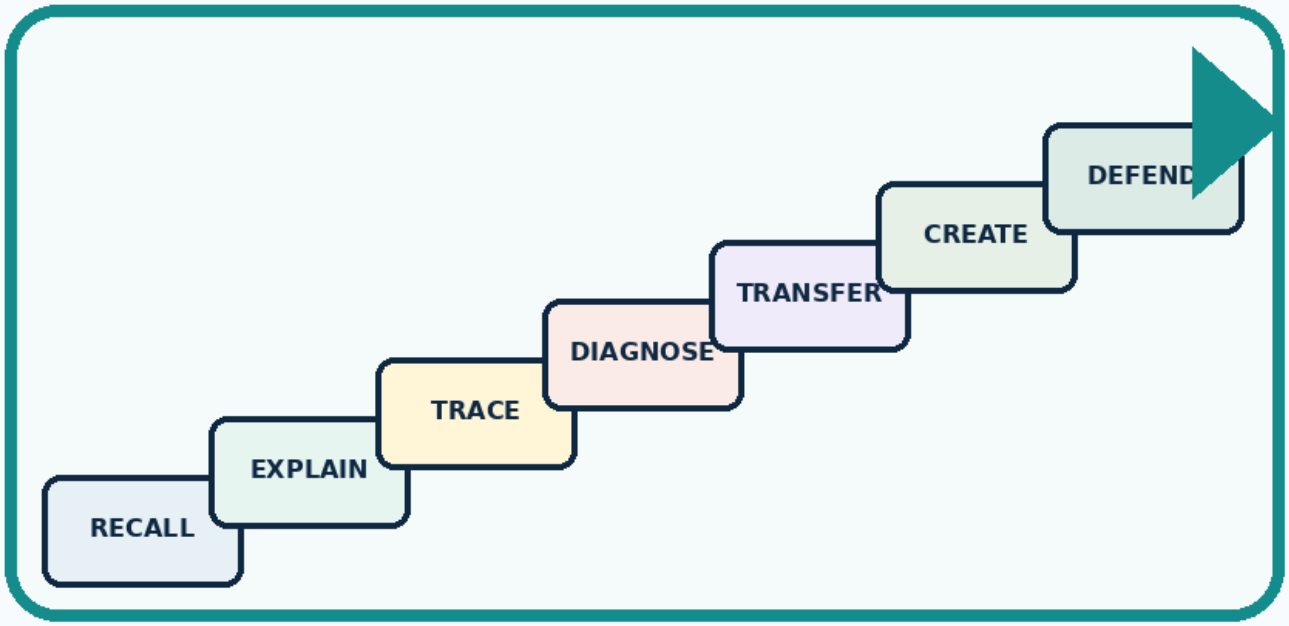
- RECALL Define **similarity**.
- EXPLAIN Recommendations often begin by comparing selected traits, not whole objects.
- TRACE Show the **input**, **process**, and **output**.
- DIAGNOSE Repair: The nearest item is automatically the best recommendation.
- TRANSFER Apply the idea to a new home or classroom case.
- CREATE Design one revealing test.
- DEFEND Give a claim, **evidence**, **mechanism**, limit, and safe action.

▼ Earlier wording, alternatives and picture transcripts

PRACTICE LADDER

Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

▼ **Reference: check your explanation**

REFERENCE · SECTION 12

Reference: check your explanation

> Use after practice. Transfer the comparison to books is the selected independent case.

Check the evidence and its limit

- ☐ Name or draw the fresh case.
- ☐ Trace **input**, **process**, and **output**.

- ☐ State the conclusion and strongest **evidence**.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Record support

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

▼ **Earlier wording, alternatives and picture transcripts**

MASTERY

Checklist for explaining this week's idea

Use after practice. Transfer the comparison to books is the selected independent case.



EXPLANATION CHECK State the candidates, selected traits, feature contributions, **weight**s and tie policy; preserve unknown values and distinguish a score from a hard suitability requirement.

Independent **evidence**:

SELF-CHECK Plot items consistently, define how each axis is measured, and compare **distance**s. A different axis or **weight** can change the nearest item without any item moving in the real world.

NEXT BRIDGE Carry today's **evidence** habit into: Recommendations.

▼ **Extra practice: Count matching book traits**

REFERENCE · SECTION 13

Extra practice: Count matching book traits

> Try the question before opening its worked explanation.

Read the supplied case and question

Book A has animals and adventure; B has animals only; C has neither. A reader wants both traits.

Rank by number of matching traits.

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

A scores 2, B 1, C 0. Ranking depends on the chosen traits; it is not a universal measure of book quality.

Evidence or stage	Value or result
Book A	2
Book B	1
Book C	0

▼ Earlier wording, alternatives and picture transcripts

Week 19 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Count matching book traits

Book A has animals and adventure; B has animals only; C has neither. A reader wants both traits.

Rank by number of matching traits.

A scores 2, B 1, C 0. Ranking depends on the chosen traits; it is not a universal measure of book quality.

▼ Reference: original lesson checklist

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Calculate similarity using explicitly selected features and a stated scoring rule	Explain the deciding step, show a trace or test, and state one limit.
Explain how a change in feature weights can change the highest scoring item	Explain the deciding step, show a trace or test, and state one limit.
Handle ties and missing features without claiming that a similarity score proves suitability	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: A fictional craft-card matcher compares each **candidate** with a target.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional craft-card matcher compares each **candidate** with a target.
- Which labelled step matches this detail? New scores A=4, B=4, C=2; A and B remain tied.
- 1. Target features
- 2. Initial result
- 3. Texture **weight** 3
- 4. Equal **weight**s
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Texture **weight** 3 New scores A=4, B=4, C=2; A and B remain tied. **Similarity** is defined by the chosen representation and **weight**s.

The cards did not change when texture gained **weight**; the comparison rule did. Two high scores need not mean two identical cards.

A complete rule also says what to do with tied scores, rather than choosing whichever appears first by accident.

Changed-case question

- Changed-case practice: Keep original **weight**s but change the target texture to rough. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. This one example guarantees correct results for every future **input**.
- 3. C scores 3; A and B each score 1.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the changed case**

After your attempt, compare with the supported explanation: C scores 3; A and B each score 1. C now matches all three target features; the others lose their smooth-texture match.

Repair and save your work

Review Confidence Is Not Correctness (ai-1/18). Goal: Distinguish a confidence claim from **evidence** that a **prediction** was correct. Short practice: Was the unknown label guaranteed correct because its confidence was **80%**?.

Return to this week and explain what you changed.

Review Boundaries and 'Not Sure' (ai-1/16). Goal: Identify a specific source of uncertainty and explain how it affects a **prediction**. Short practice: Which original block needs a better width before accepting or rejecting it?

What should the reviewer do?. Return to this week and explain what you changed.

Review Learning from Examples (ai-1/13). Goal: Fit a threshold by tracing **prediction**s and counting disagreements for every **candidate**. Short practice: Which new measured length would reveal disagreement between the **candidate**s?.

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

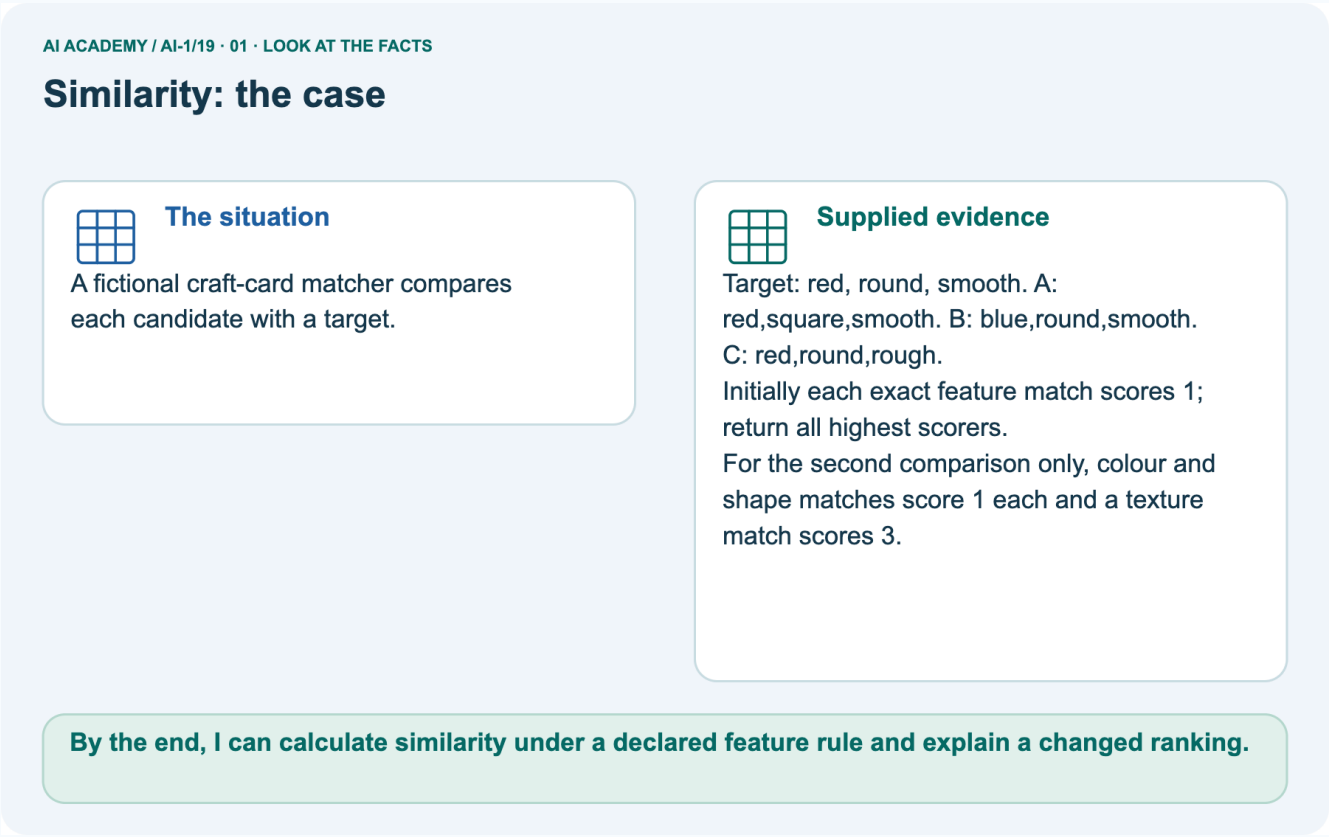
Keep your completed work

▼ Reference: the six-picture journey

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. Similarity: the case



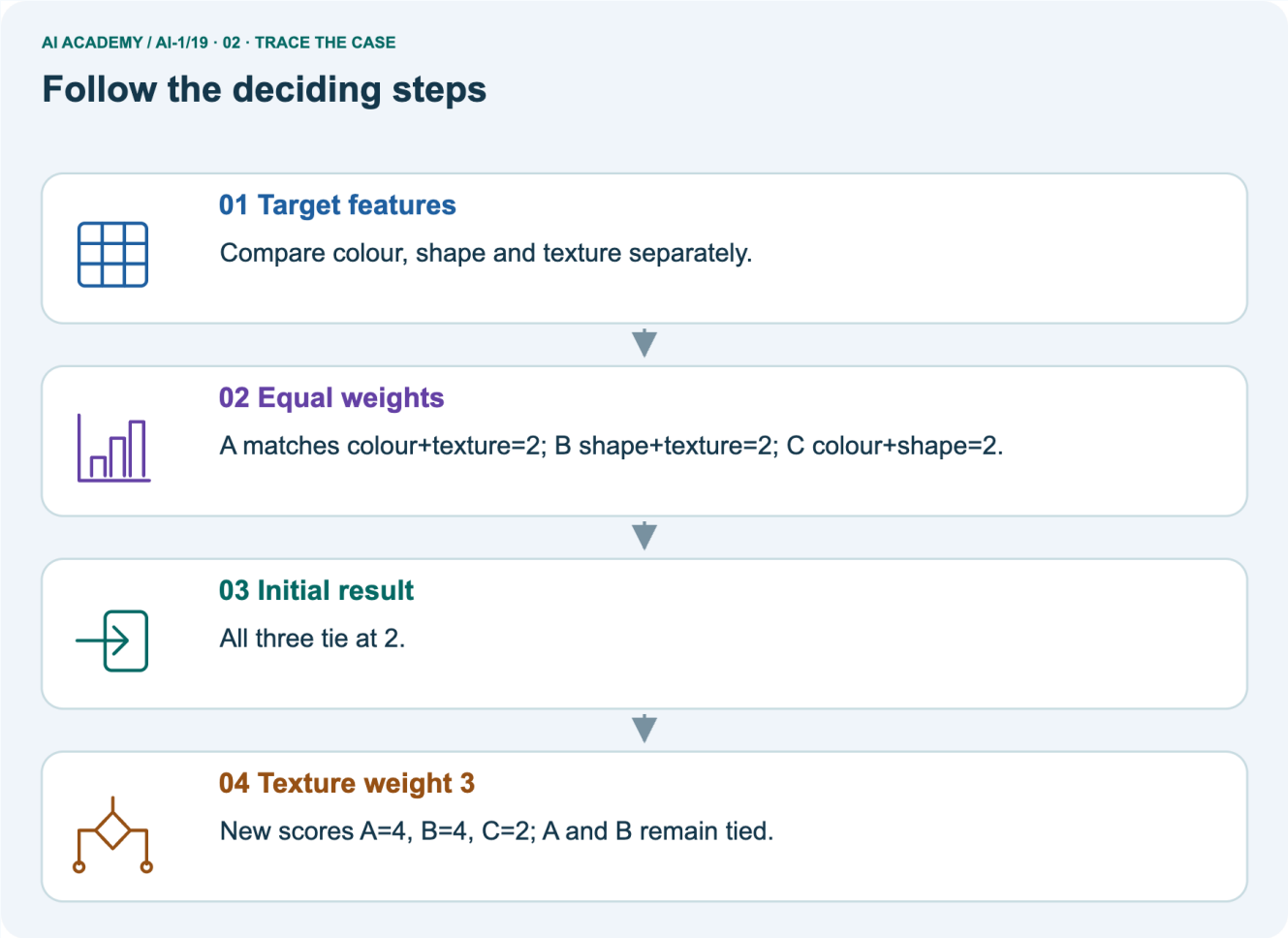
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional craft-card matcher compares each **candidate** with a target. Supplied facts: Target: red, round, smooth. A: red,square,smooth. B: blue,round,smooth. C: red,round,rough.

Initially each exact **feature match** scores 1; return all highest scorers. For the second comparison only, colour and shape matches score 1 each and a texture match scores 3.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

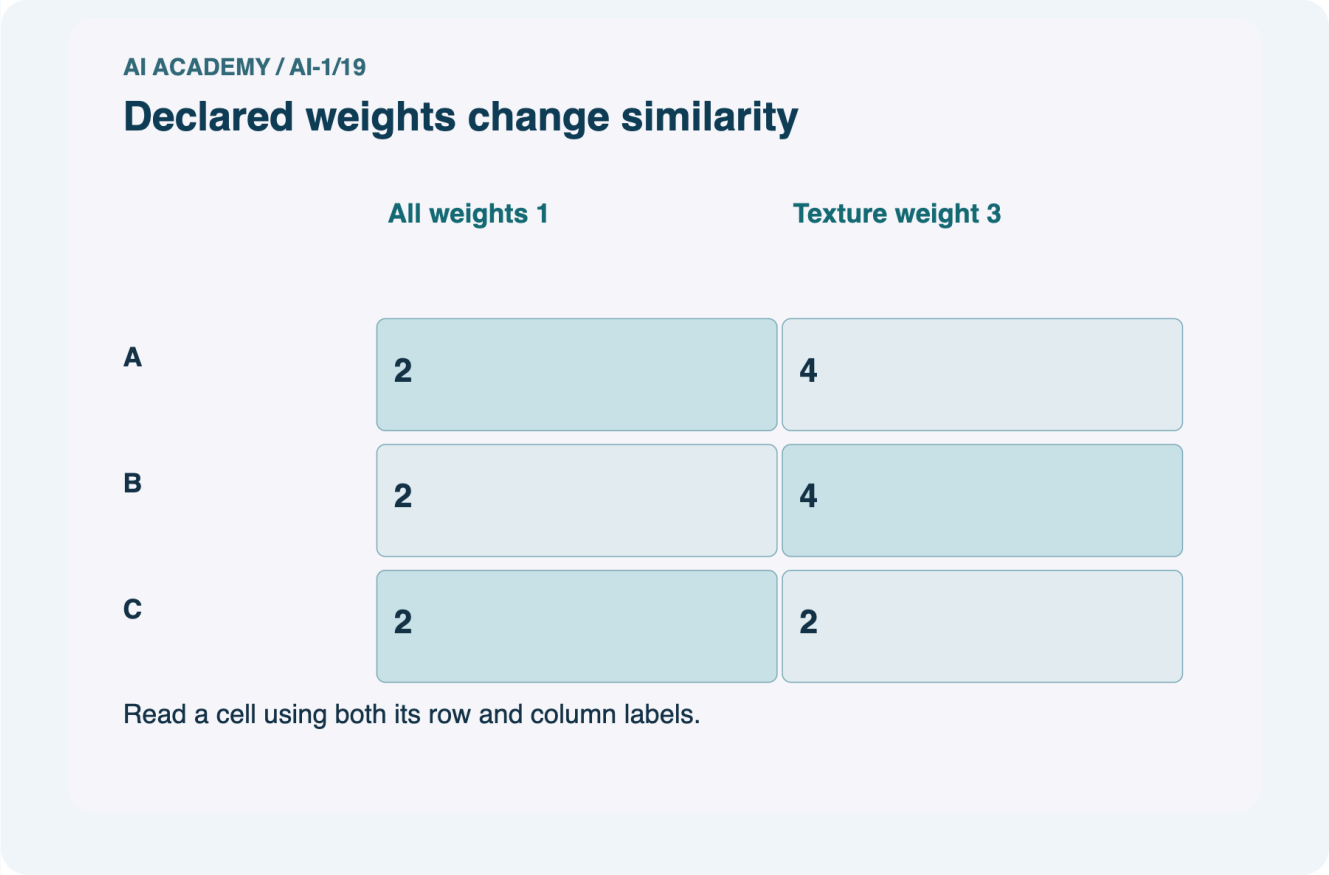


Follow the labelled picture with the matching explanation. Open the vector diagram

Target features: Compare colour, shape and texture separately. → Equal **weight**s: A matches colour+texture=2; B shape+texture=2; C colour+shape=2. → Initial result: All three tie at 2. → Texture **weight** 3: New scores A=4, B=4, C=2; A and B remain tied.

Point to each numbered stage and explain which recorded **input** it uses.

03. Declared weights change similarity

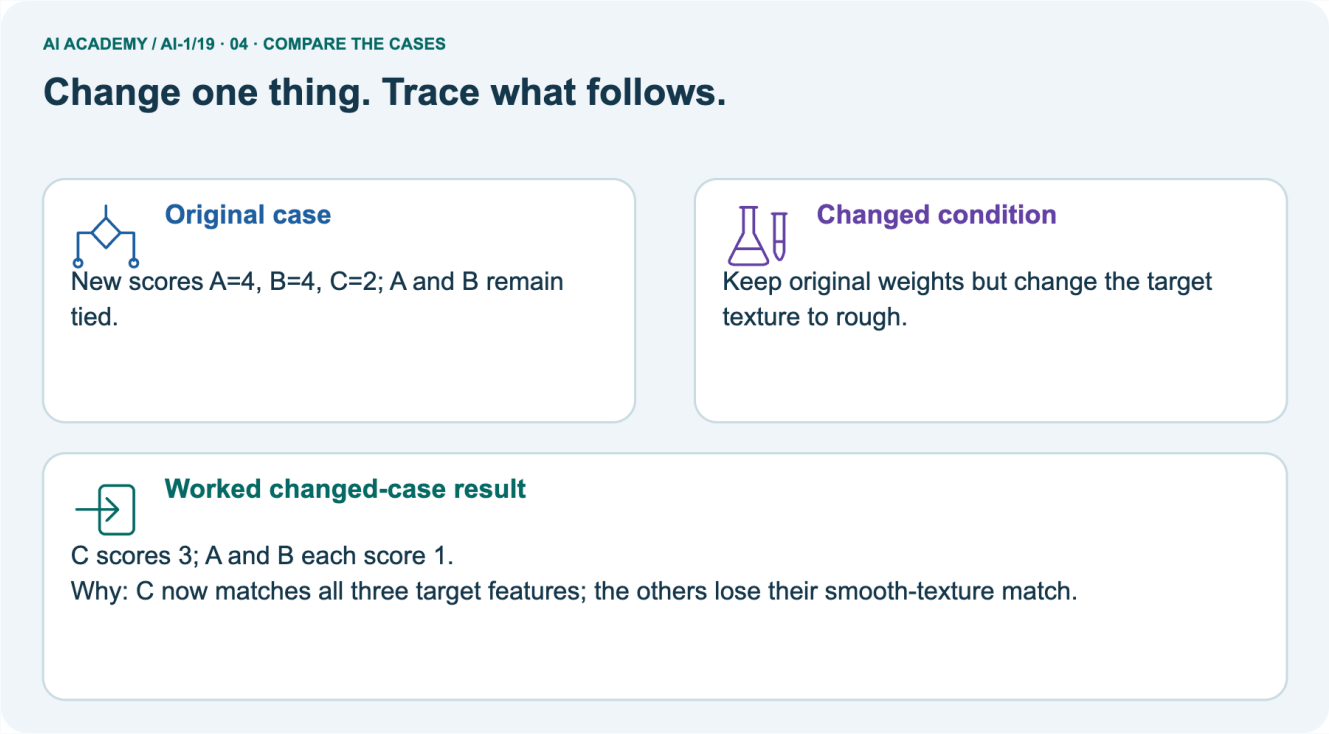


Follow the labelled picture with the matching explanation. Open the vector diagram

Declared **weight**s change **similarity**. A: All **weight**s 1 = 2, Texture **weight** 3 = 4. B: All **weight**s 1 = 2, Texture **weight** 3 = 4. C: All **weight**s 1 = 2, Texture **weight** 3 = 2.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



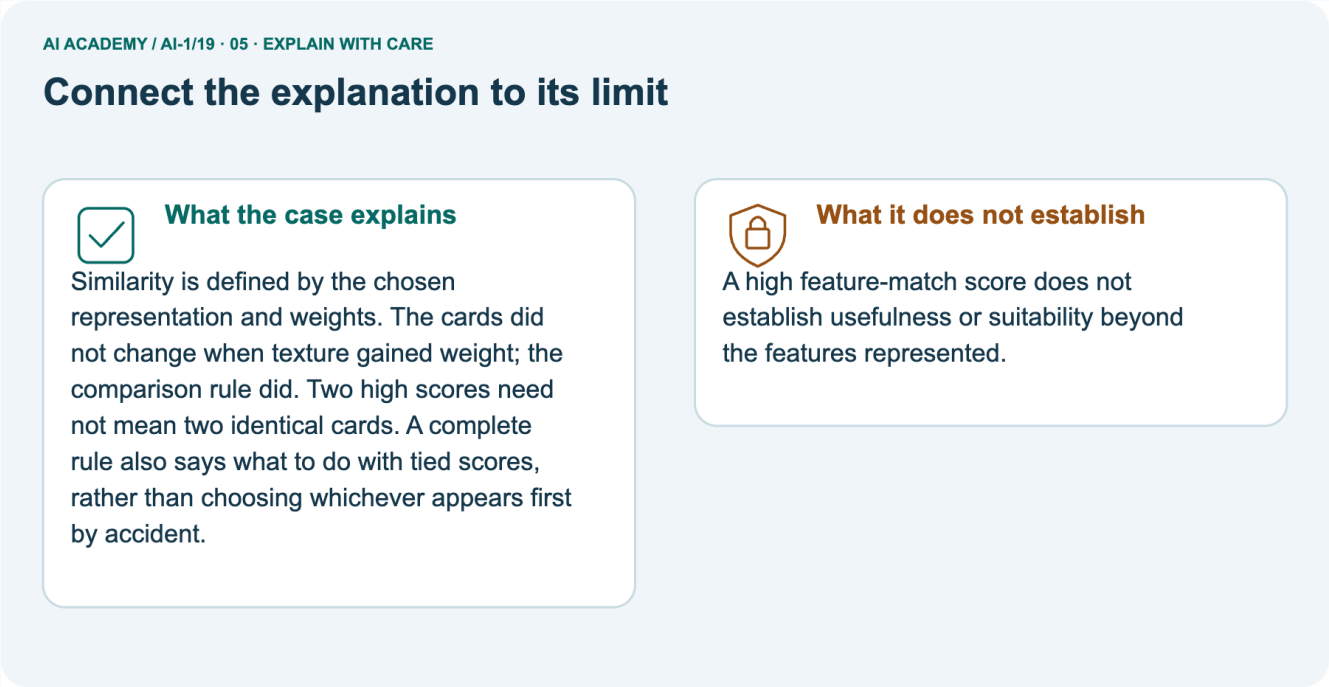
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: New scores A=4, B=4, C=2; A and B remain tied. Changed condition: Keep original **weight**s but change the target texture to rough. Expected result: C scores 3; A and B each score 1.

Reason: C now matches all three target features; the others lose their smooth-texture match.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



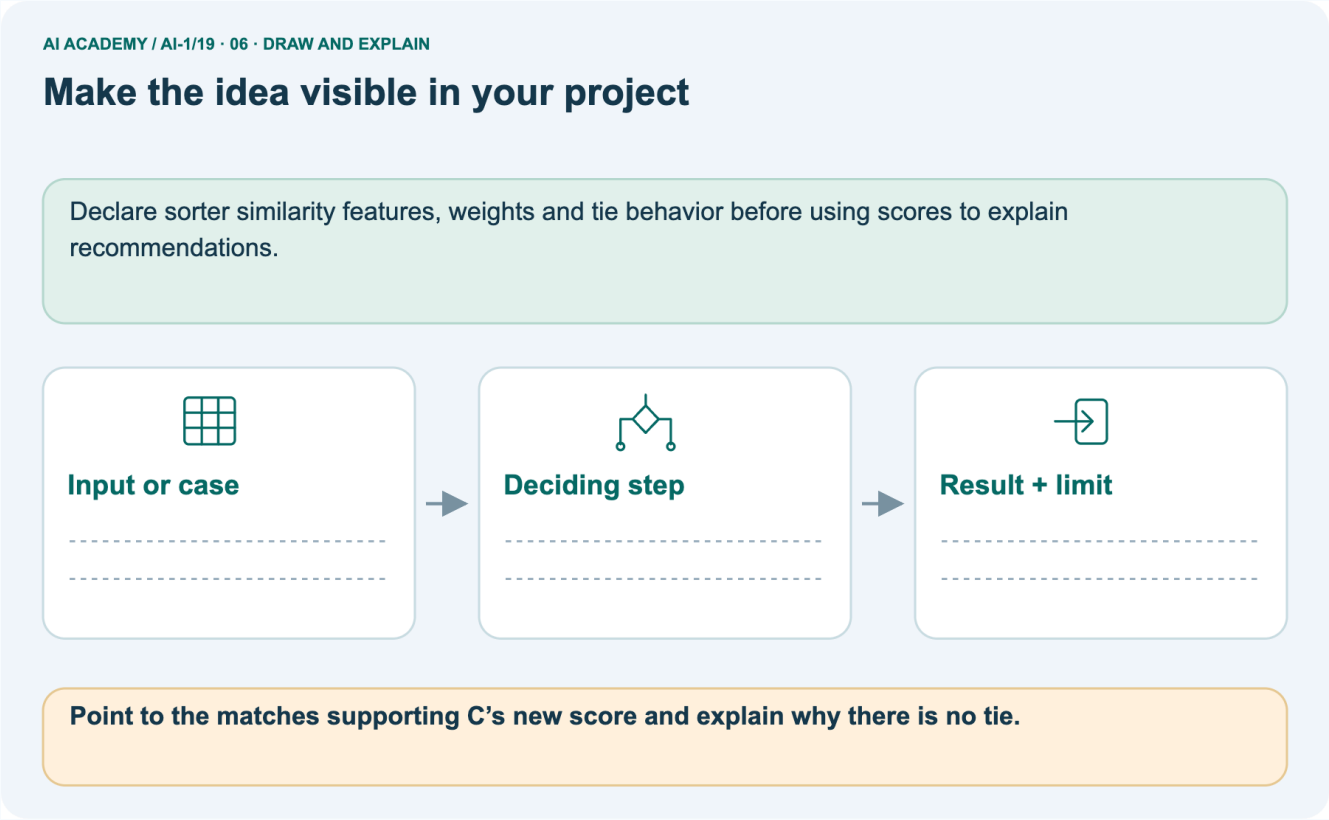
Follow the labelled picture with the matching explanation. Open the vector diagram

Similarity is defined by the chosen representation and **weight**s. The cards did not change when texture gained **weight**; the comparison rule did. Two high scores need not mean two identical cards.

A complete rule also says what to do with tied scores, rather than choosing whichever appears first by accident. Limit: A high feature-match score does not establish usefulness or suitability beyond the features represented.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Declare sorter **similarity** features, **weight**s and tie behavior before using scores to explain recommendations. Draw your **input**, deciding step, result and limit.

Supported question: Point to the matches supporting C's new score and explain why there is no tie.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Similarity: the case

02. Follow the deciding steps

03. Declared weights change similarity

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.